EXPT NO: 10	PERT -ANALYSIS
DATE:	

AIM:

To perform Program Evaluation Review Technique (PERT) Analysis for project scheduling and management using Python.

PROCEDURE:

1. **Install Required Libraries:**
o Make sure you have pandas, networkx, matplotlib, and numpy installed. You can install them using pip if not already installed.
bash
Copy code
pip install pandas networkx matplotlib numpy

2. Define the Problem:
o Identify the tasks, their optimistic, most likely, and pessimistic durations, and dependencies.

3. Set Up the Problem in Python:
o Use NetworkX to represent the project tasks and their dependencies.
o Calculate the expected duration and variance for each task.
o Use topological sort to identify the critical path and perform PERT analysis.

4. Run the Code:
o Execute the Python code to get the critical path, project duration, and associated uncertainty.

5. Analyze the Output:
o Interpret the results and validate the solution against the project schedule.

PROGRAM:

```
import numpy as np
import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt
tasks = {
    'A': {'optimistic': 2, 'most_likely': 4, 'pessimistic': 6, 'dependencies': []},
    'B': {'optimistic': 1, 'most_likely': 2, 'pessimistic': 3, 'dependencies': ['A']},
    'C': {'optimistic': 1, 'most_likely': 3, 'pessimistic': 5, 'dependencies': ['A']},
    'D': {'optimistic': 2, 'most_likely': 2, 'pessimistic': 4, 'dependencies': ['B']},
    'E': {'optimistic': 3, 'most_likely': 5, 'pessimistic': 7, 'dependencies': ['C']},
    'F': {'optimistic': 1, 'most_likely': 4, 'pessimistic': 6, 'dependencies': ['D', 'E']}
}
for task, data in tasks.items():
    data['expected_duration'] = (
        data['optimistic'] + 4 * data['most_likely'] + data['pessimistic']
    ) / 6
    data['variance'] = (
        (data['pessimistic'] - data['optimistic']) / 6
    ) ** 2
G = nx.DiGraph()
for task, data in tasks.items():
    G.add_node(task,
               duration=data['expected_duration'],
               variance=data['variance'])
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    for dep in data['dependencies']:
        G.add_edge(dep, task)

topological_order = list(nx.topological_sort(G))
earliest_start = {task: 0 for task in topological_order}
earliest_finish = {task: 0 for task in topological_order}
for task in topological_order:
    earliest_finish[task] = (
        earliest_start[task] + G.nodes[task]['duration']
    )
    for successor in G.successors(task):
        earliest_start[successor] = max(
            earliest_start[successor],
            earliest_finish[task]
        )

latest_finish = {task: earliest_finish[task] for task in topological_order}
latest_start = {task: earliest_start[task] for task in topological_order}
for task in reversed(topological_order):
    for predecessor in G.predecessors(task):
        latest_finish[predecessor] = min(
            latest_finish[predecessor],
            latest_start[task]
        )
        latest_start[predecessor] = (
            latest_finish[predecessor] - G.nodes[predecessor]['duration']
```

```
        )
critical_path = [
    task for task in topological_order
    if earliest_start[task] == latest_start[task]
]
project_duration = earliest_finish[topological_order[-1]]
project_variance = sum(
    G.nodes[task]['variance'] for task in critical_path
)
project_std_dev = np.sqrt(project_variance)
print("Earliest start times:", earliest_start)
print("Earliest finish times:", earliest_finish)
print("Latest start times:", latest_start)
print("Latest finish times:", latest_finish)
print("Critical path:", critical_path)
print("Project duration (expected):", project_duration)
print("Project standard deviation:", project_std_dev)
pos = nx.spring_layout(G)
nx.draw(G, pos, with_labels=True, node_size=3000, node_color='lightblue')
nx.draw_networkx_edge_labels(
    G, pos,
    edge_labels={(u, v): f"{u}->{v}" for u, v in G.edges}
)
plt.show()
```

OUTPUT:

Earliest start times: {'A': 0, 'B': 4, 'C': 4, 'D': 6, 'E': 7, 'F': 10}
Earliest finish times: {'A': 4, 'B': 6, 'C': 7, 'D': 8, 'E': 12, 'F': 14}

Latest start times: {'A': 0, 'B': 4, 'C': 4, 'D': 6, 'E': 7, 'F': 10}
Latest finish times: {'A': 4, 'B': 6, 'C': 7, 'D': 8, 'E': 12, 'F': 14}

Critical path: ['A', 'C', 'E', 'F']

Project duration (expected): 14.0
Project standard deviation: 1.247219128924647

RESULT:

The PERT – ANALYSIS has been completed successfully.